

[Home](#) › [My Courses](#) › [Financial Markets](#) ›
[M6: Model Failure and Crises](#) › Lesson Notes

Lesson Notes

MODULE 6 | LESSON 2

DEBT AND EQUITY FINANCING: HOUSING DEVELOPMENT EXAMPLE

Reading Time	45 minutes
Prior Knowledge	Balance sheet, Stock, Bonds
Keywords	Urbanization, Equity financing, Debt financing, Real estate investment Trust (REIT), Bank loan, Cooperative development, Private equity, Real estate developer, Landlord

In the last module, we discussed the main components of the balance sheet (assets, debt, and equity) and how these translate into a mortgagor's leverage ratio, which in turn correlates positively with the probability of default. We also looked at the typical elements of a bank's balance sheet (e.g., reserves and loans on the assets side, deposits and borrowings on the liabilities side). Banks are usually the lender in a mortgage transaction, so we have a better understanding of the real estate and mortgage market when we know how their balance sheet drives profitability by way of the "gross spread."

We also saw in the last module that houses are not bought and sold in a vacuum: A mortgage loan is not just a document that ties borrowers to their lending bank; rather, a single mortgage loan can ultimately involve an arranger, an investment bank, a servicer, a guarantor, various investors with different appetites

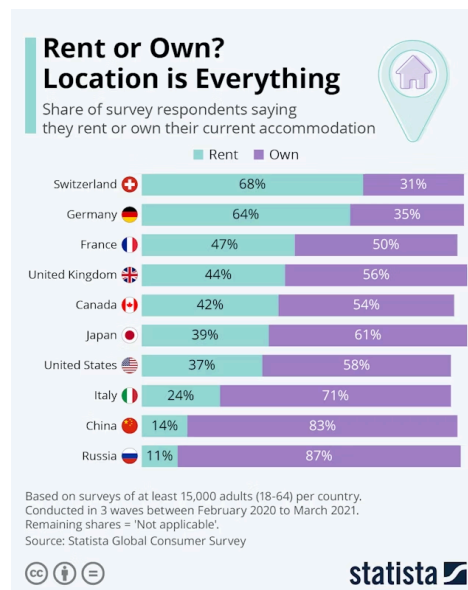
for risk, and many other market participants. We have considered the credit risk of both the individual mortgage borrower as well as the entire mortgage collateral pool. And yet there are still so many other financial decisions and market players vital to the real estate market.

1. To Rent or to Own?

Houses can be seen as investments, although with a few caveats: houses need maintenance, houses may need expensive fixes, housing markets are usually local, and if the local economy encounters headwinds, the value of houses could suffer as well. However, it is pretty clear that many people see houses as investments. The graphs on the website of the Federal Reserve Bank of Saint Louis contain the time series related to house pricing [FRED_Housing_Data](#). If you are interested in a more in-depth look into U.S. housing, you can find a lot of information on [FRED_Overall_Housing_Data](#) and [US_Housing_Data](#). We could also compare investing in the stock market with investing in real estate, but it remains that housing is an important investment.

We have seen how individual mortgages can be approved and securitized. The financial analyses involved in each process can be complex, and yet these are just two facets of housing finance. Before anything can happen with the mortgage, an individual or household must first decide whether to buy, rather than rent, a home. Depending on what country you live in (among many other factors), it may be the norm to do one or the other.

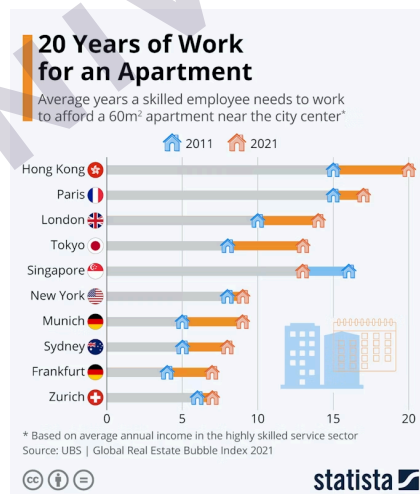
Figure 1: Renting vs. Owning Around the World



Source: Armstrong, Martin. "Rent or Own? Location is Everything." *Statista*, 27 April 2021. <https://www.statista.com/chart/24738/gcs-share-renting-and-owning-home/>.

Besides cultural norms, real estate can be expensive to buy (and sometimes to rent as well). Check out this graphic showing how long a resident in each of these countries must work to afford a home, and in almost all places, that length of time is getting longer each year.

Figure 2: How Long Before I Can Afford an Apartment?



Source: Armstrong, Martin. "20 Years of Work for an Apartment." *Statista*, 29 Oct. 2021. <https://www.statista.com/chart/26082/years-needed-to-buy-an-appartment/>.

Individuals seeking to optimize their financial situation need to ask themselves, "Should I rent or should I own my home? When is it optimal to do one or the other?" [This first required reading video](#) provides a draft framework for making this decision: as Khan makes clear, all the inputs (and even the types of inputs) to this model depend entirely on the individual's unique circumstances, just as the astute analysis of any investment or

corporate project depends on the particulars of the financial undertaking under consideration. No single model or set of inputs will suffice in every situation.

As you watch the comparative analysis between renting and buying a home, keep track of the expenses and the savings associated with each choice. If you know anything about the housing market where you live, you might mentally add or subtract items to these lists to model your own potential decision. Be sure to also note the so-called *intangibles* or *intangible assets*. Intangible literally refers to things that you cannot touch; intangible assets cannot be touched, and they can often be difficult to value or price: stability and flexibility, for example, are intangible, but as discussed in the video, they have subjective values that should also be included in the decision model.

2. Real Estate Market: Developers and Landlords

Just as there is no mortgage unless someone wants to buy a house instead of renting one, one cannot decide to buy a house that does not exist in the first place. Real estate developers are thus another crucial participant in the real estate market. What are real estate developers? Are they the same as landowners and/or landlords? The title of the second required reading (Herriges) gives away the answer to this question. A key takeaway from this reading is the following:

“A landlord profits by **possessing** real estate. A developer profits by **creating** real estate.”

But the article provides more information about the implications and subtleties of this distinction, which should be kept in mind when considering real estate investment. For this reading, pay attention to the factors driving the profit for each of the market players discussed: which entities profit from the scarcity in a housing market or from rising real estate prices? In what ways do landowners and real estate developers have the same interests, and in what ways do they have different interests? In what ways do landlords and homeowners have the same interest?

3. The Costs of Development and How to Finance Them

While the second reading touches on the developer costs, the third required reading, "6 Things to Know about How Development Works", provides a bit more background about the developer's business model, especially the costs and financing constraints. For this reading, pay attention to the developer's main expenses and how these are financed. (The accompanying graphics display these nicely.)

After learning about the balance sheet in the last lesson (specifically, how assets must "balance" with the sum of equity and liabilities), it should come as no surprise that the developer's financing generally takes two forms: debt (i.e., liabilities) and equity. Developers can finance an increase in assets with different types of debt and equity. The framework for deciding how to finance a real estate development can be adapted to the financing decisions for many corporate finance projects and is not all that different from the financing decision that an individual makes with respect to their use of debt (e.g., a mortgage) on their personal balance sheet.

Wu describes five financing channels for developers, four of which should be familiar to you after the last lesson's discussion of the balance sheet (1339-40).

3.1 Liabilities

Bonds: The developer can issue bonds, effectively borrowing from a lender in the credit market, as discussed in Module 5. Bonds can have short and long tenors, but both have their disadvantages: a short tenor means the principal must be repaid soon. The debtor firm may be able to repay the bonds with the proceeds from a new bond issuance in the future, but that is risky as the conditions and circumstances may change by then. The disadvantage of a longer tenor is that these bonds have higher coupon (interest) rates, which makes them more expensive for the borrower.

Bank loans: The developer can borrow money from a bank instead of the credit market.

These are both debt (or liability) investments. Bonds can trade from investor to investor, similar to stock. Bank loans, on the other hand, are much less liquid, meaning they are not as easily

traded. Białek-Jaworska and Krawczyk also mention that bank loans are more expensive and more restrictive (264) for the borrower.

3.2 Equity

Stock market: The developer can sell an ownership stake in itself (the development company) by issuing stock that trades on a stock exchange. Issuing stock can be difficult and complicated.

Real estate investment trust (REIT): The developer can sell an equity stake in itself to an REIT, a company that only invests in real estate.

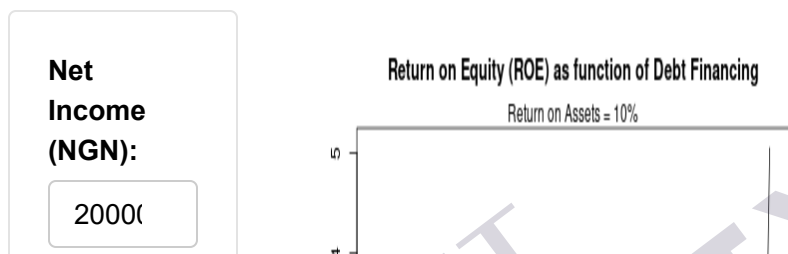
Both of these financing options involve increasing the equity amount on the balance sheet. By selling additional equity in the company, the total equity increases, which allows for additional assets. The additional assets are financed from equity investment. Stock that trades on the stock market is called “publicly traded.” On the other hand, the developer could sell equity to an REIT in a private transaction; such equity is therefore called private equity.

As Wu states, equity financing has advantages and disadvantages: selling a stake in the company means sharing the risk, as financial losses would be shared by the additional equity holders, but it also means sharing the profit with additional parties (1339). The other advantages of equity financing are that 1.) there is no “debt service” (interest payments), and 2.) the capital is basically permanent (Wu 1339).

Cooperative development: In the case of “cooperative development,” the developer does not actually increase its own assets, so there is no increase in debt or equity per se. This special arrangement allows the developer to partner with another firm, like with a construction company to reduce or eliminate construction costs or with a landowner to reduce or eliminate the capital required to buy the land to be developed. With this financing option, the developer does not sell any of its own equity, but it does have a profit-sharing agreement with its partner(s). Nevertheless, this approach can be seen as selling equity in a particular development project as opposed to in the developer itself.

Reading about the differences between debt and equity financing is one thing. Let's use this example to demonstrate the impact on leverage and shareholder profits resulting from the financing mix.

Return on Equity as a Function of Capital Structure and Interest Rates



4. Conclusion

In this lesson, we revisited personal finance and sketched up a framework for deciding whether to buy or rent a home. We also looked at two more participants in the real estate market: the landlord and the developer. We took a closer look at the developer to see financing in action. An increase in assets requires either an increase in equity or an increase in debt (or both), so the developer has to make a careful, sometimes difficult, decision about sharing risk, profit, and ownership of a development. The real estate developer's debt and equity is completely analogous to the debt and equity of an individual. As we saw in the last lesson, individuals also have debt (for example, a mortgage, a home equity loan, an auto loan, and/or a personal loan), and they have equity (the amount of the excess of assets over these liabilities).

As we will see in the following lesson, debt and equity investment can be combined in various innovative ways, with varying degrees of success, in different housing markets. We will also see how real estate investments have evolved over the past few decades and in different ways in different parts of the world.

References

- Armstrong, Martin. "20 Years of Work for an Apartment." *Statista*, 29 Oct. 2021.

<https://www.statista.com/chart/26082/years-needed-to-buy-an-apartment/>.

- Armstrong, Martin. "Rent or Own? Location is Everything." *Statista*, 27 April 2021.
<https://www.statista.com/chart/24738/gcs-share-renting-and-owning-home/>.
- Białek-Jaworska, Anna, and Tomasz Krawczyk. "Corporate Bonds or Bank Loans? The Choice of Funding Sources and Information Disclosure of Polish Listed Companies." *Central European Economic Journal*, vol. 6, no. 53, 2019, pp. 262-285.
<https://doi.org/10.2478/ceej-2019-0017>.
- Herriges, Daniel. "Landlords Are Not Developers (and Vice Versa)." *Strong Towns*,
<https://www.strongtowns.org/journal/2021/9/2/landlords-are-not-developers-and-vice-versa>.
- Herriges, Daniel. "6 Things to Know About How Development Works." *Strong Towns*,
<https://www.strongtowns.org/journal/2020/9/15/6-things-to-know-about-development>.
- "Renting versus Buying a Home." *Khan Academy*,
<https://www.khanacademy.org/economics-finance-domain/core-finance/housing/renting-v-buying/v/renting-versus-buying-a-home>.
- Wu, Tingting. "Analysis of Financing Plan for Real Estate Development Project." *Open Journal of Business and Management*, vol. 7, no. 3, 2019, pp. 1338-1345.
[10.4236/ojbm.2019.73092](https://doi.org/10.4236/ojbm.2019.73092).